

KOGI

Independent Worker Operating System

Unified Work Management Platform for the Independent Economy

Platform Design Document

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Kogi Team

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1. Platform Overview

1.1 Vision & Mission

Kogi is the Independent Worker Operating System — a unified, configurable, and community-connected work management platform designed for the modern independent economy. Kogi unifies the tools and systems that workers use to execute their processes, generate and consume value, and connect with one another into a single scalable platform.

The platform provides independent contractors, freelancers, gig workers, entrepreneurs, creatives, and cooperative organizations with the infrastructure they have historically had to assemble piecemeal: project management, financial accounts, marketplace access, community, benefits, and AI assistance — all tracked, governed, and maintained through a master hyperspreadsheet.

Design Principle: Everything in the Kogi ecosystem is a Portfolio Item. The Portfolio System is therefore the universal abstraction layer that every platform application builds upon.

1.2 Core Platform Architecture

The Kogi Platform has eight core modules that together constitute a complete independent worker operating system:

Module	Code	Primary Responsibility	Sub-Modules / Key Features
Dashboard	KDSH	High-level portfolio OS overview; independent worker desktop screen	Portable Benefits · Initiatives · Morning Briefing · OKR Progress
Portfolio	KPRT	Complete work lifecycle management; master hyperspreadsheet	Programs · Projects · Assets · Artifacts · Resources · Tools · Containers
Wallet / Bank	KBNK	Financial operating layer; accounts, ledgers, payments, escrow	Accounts · Wallets · Ledger · Escrow · Campaigns · Invoicing · Tax
Office	KOFF	Portfolio item execution and management system	WMS · RMS · Booking · CRM · Contracts · Logistics
Marketplace	KMKT	Portfolio item markets, exchanges, distribution, supply chain	Labor Market · Asset Store · Exchange · Deal Room · Campaigns
Spaces	KSPC	Portfolio space and community management system	Feed · Rooms · Events · Organizations · Social Graph · Notifications
Hub	KHUB	Multi-portfolio governance management system	Shared Portfolios · Governance · Voting · Rights Management · Federation

Module	Code	Primary Responsibility	Sub-Modules / Key Features
Agent	KAGT	Portfolio automations; AI assistant system	Oba AI · Multi-agent orchestration · Workflow automation · Analytics

1.3 Platform Substrate — Hypergrid

Kogi is built on Hypergrid, an N-dimensional distributed spreadsheet engine. Every entity in the platform is a row in a Hypercube, every field is a cell at a typed coordinate, and the entire system is governed by CRDTs (Conflict-free Replicated Data Types), an append-only EventLog, a Hypergraph relational layer, and an AI writeback pipeline.

The primary Hypercube for Kogi is `kogi.portfolio.components`, which holds every entity in a worker's professional life — programs, projects, tasks, artifacts, assets, gigs, benefits, and financial records — as HyperRows in a single N=2 cube. The 35-sheet system is 35 saved HypercubeViews over the same underlying data.

Substrate Component	Description
Universal Cell Store	N-dimensional coordinate space with per-attribute CRDT semantics. D_1 = entity axis (UUID), D_2 = property axis (field name), D_3+ add time, segment, or other dimensions.
Hypergraph	Relational layer. Every HyperRow is a HypergraphNode. Typed edges (Hierarchy, Dependency, Association, CrossGridLink, Employs, Collaborates) connect entities within and across Grids.
EventLog	Append-only, immutable, time-travelable. Every mutation is recorded with actor, VectorClock, and payload. AS_OF operator replays to any prior timestamp.
CRDT Engine	Per-attribute CRDT semantics: LastWriteWins for scalars, OR-Set for set fields, Lattice for lifecycle states, PNCounter for transactional accumulators, MaxRegister for versions.
AI Writeback Pipeline	kogi-engine subscribes to EventLog mutations, computes domain signals (health scores, risk scores, anomaly flags), and writes results back as system-only AI computed attributes.
HyperQL	SQL-plus query language with N-dimensional cell references, DimSlice, DimFold, ROLLUP, DRILLDOWN, AS_OF time-travel, TRAVERSE GRAPH, and IDENTITY scoping.
KLNK (Kogi Link Network)	The economic graph of interconnected worker portfolios. CrossGridLink edges with ShadowCells model employment, investment, collaboration, and marketplace transactions.

2. Core Personas

2.1 Persona Overview

Kogi serves a diverse range of independent worker archetypes. Personas are labels associated with users to provide customized experiences and help other users find and filter participants by type. A single user may carry multiple personas simultaneously.

2.1.1 The Freelancer

Entrepreneurs, micropreneurs, gig workers, freelancers, contract workers, consultants, coaches, service-based workers, developers, and side hustlers. This is the primary platform persona — workers who need a systematic work management system and resources to build and develop their personal work portfolios.

2.1.2 The Creative

Content creators, musicians, artists, writers, journalists, media personalities, podcasters, and influencers. Creators looking for a central system to track, manage, and showcase their work, and to connect with other creatives and with those who may desire their services.

2.1.3 The Enthusiast

Hobbyists, spectators, donors, investors, venture capitalists, and skilled workers looking for a new challenge. People looking to invest in things that interest them — video games, new products, interesting service offerings. Enthusiast portfolios tend to contain items they have invested in or contributed to, rather than projects they are directly working on.

2.1.4 The Professional

Professional services workers in a '9-to-5' who want a work system to manage their jobs — the 'employed entrepreneur' or 'entrepreneurial employee.' Legal services, accountants, professional service consultants, engineers, and medical professionals. Veterans who already have mature bespoke work systems in place; the platform facilitates and organizes their assistants and agents to manage their systems, and allows them to market and promote themselves.

2.1.5 The Cooperative

Organizations, teams, collectives, federations, autonomous organizations, and cells. Multi-participant entities that need shared governance, collective treasury, revenue distribution, and decentralized work portfolio management.

2.1.6 The Explorer

People just checking out the platform — looking for community, resources, time, and money. Cold-start users with minimal interaction history; the platform defaults to trending and popular content for this persona.

2.2 Macro / Micro Persona Dimensions

Dimension	Variants
Technical sophistication	Technical users · Non-technical users · Power users

Dimension	Variants
Engagement depth	Power users (high frequency, advanced features) · Casual browsers (low engagement, trending content) · Specialists (deep domain focus) · Collaborators (high social engagement)
Economic role	Sellers of labor/assets · Buyers/hirers · Investors/donors · Multi-role participants
Organization type	Solo worker · Team member · Cooperative member · Collective participant · Federation member

3. Portfolio System

3.1 Overview

The Portfolio is the root domain concept of Kogi. It is the complete, unified record of an independent worker's professional and economic existence. Every entity in the worker's operational life is a row in the Portfolio System. The Portfolio System manages the complete work lifecycle — from produced idea to delivered consumable value and back again — in a closed-loop producer and consumer independent worker economy.

All portfolio items are contained within and managed by a master hyperspreadsheet. Users simultaneously produce and consume work items to supply and deplete their portfolios of value and work.

Portfolio Root Concept: The Portfolio IS the system. Every row in the master hyperspreadsheet IS a portfolio component. The spreadsheet IS the operating system.

3.2 Portfolio Structure Hierarchy

The portfolio structure follows this hierarchy from broadest to most granular:

Level	Description
Portfolio	Top-level container organizing all of a user's or organization's work, assets, and goals. May contain sub-portfolios, programs, and projects. Functions as a content management system.
Program	Multiple projects aligned to achieve common outcomes and objectives; shares milestones across constituent projects.
Project	A temporary endeavor of work to achieve a finite, distinct outcome (state change / transform). Produces assets and artifacts.
Activity	Work performed within a project — a set of coordinated tasks and events contributing to project progress.
Task (WBS) / Event (TBS)	The atomic execution unit. Tasks are work breakdown items; Events are time breakdown items anchored to the calendar.

3.3 Portfolio Item Types

All portfolio items carry RBAC permission models (Public, Protected, Private, Hidden) and are tradeable, configurable, deployable, distributable, and extensible. Users can perform a rich set of actions on any item: like, comment, share, subscribe, watch, bookmark, trade, book, donate, invest, fund, collaborate, contribute, and configure.

3.3.1 Assets

Capability-enabling items which can be consumed for future work pipelines. Transformative intermediary items between raw inputs and final delivered outputs.

- Skills, knowledge, information
- Software, devices, agents, machines
- Equity, securities, liquidity
- Portable benefits
- Funding, donations, investments, accounts, wallets
- Offers, deals, bids, proposals, requests, bookings, consultations, contracts, gigs, jobs
- Ideas, concepts, prototypes, MVPs, designs, mockups, wireframes, renderings, schematics, models, simulations
- IP, rights, licenses, patents, marks, agreements, contracts, memos

3.3.2 Artifacts

Delivered, tangible outcome items. Produced, refined output items representing completed work products. These are the deliverables a worker produces and can showcase or sell.

3.3.3 Solutions

Final delivered product and artifacts sets, deployed and distributed 'in the wild.' Delivered packages, goods, products, and services that constitute the completed output of a portfolio project.

3.3.4 Resources

Raw, consumable input items. Land, labor, capital, estate, real estate, people, skills, knowledge, information, contacts, agents, liquidity, securities, and produced artifacts/solutions ready for consumption. This supports the platform's closed-loop reusability, renewability, recyclability, and sustainability model.

3.3.5 Entities

Teams, groups, legal entities, organizations (autonomous, decentralized), communities, agents, and executors. Multi-participant organizational structures operating within the portfolio system.

3.3.6 Links (LinkTrees)

Link Structure	Description
Linktree	A user's 'personal work IP address/router' — the interface for all external communications routing, accounts across external platforms, integration toolchains, and RBAC connections.
Link Branches	Professional branches · Developer branches · Public branches · Private branches · Custom branches
Link Leaves	Specific integration endpoints — connections to individual external platforms and services.
Linkforests	Collections of linktrees across multiple identities or organizational contexts.
Linknetworks	Interconnected linkforests forming the economic graph of the KLNK system.

3.3.7 Tools

Portfolio items used to perform and execute work and tasks. Organized hierarchically:

- Toolbox — the top-level tool container for a user's work environment
- Toolset — a categorized collection of related tools within a toolbox
- Toolkit — a bundled set of tools for a specific use case or workflow
- Toolchain — a sequential series of tools executing in an automated pipeline
- Tool — individual integration or platform capability

3.3.8 Studio

Independent worker preference configuration items. The Studio captures a user's personal work environment settings, aesthetic preferences, creative space configurations, and custom work system parameters.

3.4 Portfolio Containers

Containers organize and structure portfolio items. They provide flexible, typed grouping and storage structures for heterogeneous portfolio elements.

Container Type	Description
Space	Bounded divisions or separations within a portfolio. Includes workspaces (spaces of execution), namespaces (named spaces), and rooms (spaces within spaces).
Binder	Can be filled with heterogeneous portfolio elements such as free-floating portfolios, programs, and projects, as well as other containers.
Guidebook	Structured collection of documentation for a portfolio item.
Playbook	Collection of developed work processes, programs, and strategies.
Notebook	Collection of long-lasting notes and letters.
Memobook	Collection of sticky-note ephemeral memos.
Resourcebook	Contains resource ledger, resource connections and integrations, vendor and supply chain information, and resource versioning history.
Schedulebook	Time and schedule book for a portfolio item.
Reportbook	Analytics, metrics, KPIs, and performance indicators.
Contactbook	List of contacts and connections including CRM profiles.
Dossier	Very detailed collection of item documents, reports, evidence, and facts.
Brief	Contains charter and high-level information about an item.
Folder	Hierarchical filesystem container; name-ordered.
Record	Ordered, sequential list of portfolio items.
Ledger	Master list of records and accounts.

Container Type	Description
Schedule	List of items; planning temporal breakdown structure.
Registry	Named index; list of associations and ownership.
Catalogue	Detailed, categorized list of items; curated, searchable, structured, and filterable.
Archive	Long-term stored and consolidated (encoded) portfolio items.
Directory	Searchable index of items.

3.5 Task Management Systems

3.5.1 Work Breakdown Structure (WBS)

The WBS decomposes work from strategic themes down to atomic tasks:

- Work Packages — top-level WBS nodes
- Themes — broadest strategic investment areas; parent of multiple initiatives
- Initiatives — coordinated sets advancing a theme or strategy
- Epics — large capabilities decomposed into stories; fits a PI or quarter
- Stories — primary delivery units; estimable, testable, completable in a sprint
- Tasks — atomic execution steps within a story

3.5.2 Temporal Breakdown Structure (TBS)

The TBS organizes work across time dimensions:

TBS Component	Description
Calendar	Grid/spatial temporal breakdown for day-level scheduling.
Timeline	Historical, provenance view of milestones and state changes.
Gantt	Detailed, operational time view with dependency-aware bar charts and critical path detection.
Schedules	Itemized time view of work assignments and resource allocations.
Roadmaps	High-level strategic view of long-horizon milestones; externally shareable.
Timeboxes	Time durations and time packets: Quarters, Program Increments, Sprints.
Events	Scheduled occurrences — meetings, bookings, appointments, reviews.

3.6 Permission Model

Portfolio items support a full RBAC permission model with the following visibility tiers:

Permission Tier	Ordinal	Visibility	Capabilities
Viewer	0	Public	Read-only, anonymous access
Subscriber	1	Public/Protected	Bookmark, save, follow, subscribe
Contributor	2	Protected	Comment, react, poll, join, submit contributions
Editor	3	Private	Edit content, bump version, manage content
Manager	4	Private	Manage settings and members, configure policies
Owner	5	Any	Full ownership privileges, all actions including transfer
Admin	6	System	Super-admin and platform-level override capabilities

3.7 CRDT Semantics by Attribute Type

Every attribute in the Portfolio System carries explicit CRDT semantics ensuring safe concurrent mutation across distributed nodes:

CRDT Type	Attribute Examples	Merge Behavior
LastWriteWins (LWW)	name, description, status, config, JSON blobs	Highest VectorClock timestamp wins; actor ID tiebreak
OR-Set	owners, tags, children, dependencies, toolbox_ids	All concurrent Adds survive; Removes target only exact unique_tag
PNCounter	budget_spent, hours_logged, allocated_units	Sum of positive/negative increments across all nodes; any sign permitted
GrowOnlyCounter	view_count, follower_count, restart_count	Sum of all increments; decrement operations rejected
MaxRegister	version, sequence_number, schema_version	Highest numeric value always wins regardless of causal ordering
Lattice	lifecycle_state, module_state, ccr_status	join() = least upper bound in partial order; never regresses without Admin override
AppendLog	activity_log, comment_thread, decision_log	All entries preserved; causally ordered by VectorClock
DeepMergeJson	plugin_configs, feature_flags, settings_tree	LWW per leaf JSON key; only keys in delta are updated

CRDT Type	Attribute Examples	Merge Behavior
System-LWW	health_score, risk_score, anomaly_flag, narrative_summary	AI-written; PermissionTier::System enforced; no human writes

4. Wallet & Bank System (kogi-bank)

4.1 Overview

kogi-bank is the financial operating layer of the Kogi Platform — the centralized banking, accounting, payment, escrow, investment, and capital management system for independent workers, autonomous organizations, collectives, cooperatives, and independent teams.

Every transaction, every account balance, every escrow lock, and every campaign contribution flows through kogi-bank. Every financial entity is a first-class Portfolio Item in the Portfolio System, giving it the same governance, versioning, and analytics infrastructure as any other platform component.

4.2 Currency & Transaction Units

The Kogi platform introduces its own transaction primitives:

Currency / Unit	Description
Zawadi	The basic unit of transaction on the Kogi platform. Can translate to and from liquidity (USD, foreign currency), equity, securities, credits, debits, taxes, debts, and digital currency.
Sundiata Coin	The principal cryptocurrency for the Kogi platform, operating on top of the Sundiata distributed ledger.
Exchangeable Resources	Liquidity · equity · securities · credits · skills · bartered items · labor · contracts · agreements · gigs · consultations · time · bookings · meetings · appointments

4.3 Account Architecture

Accounts are stores of capital, resources, liquidity, or equity. Wallets are the transaction surfaces — the active interfaces through which financial items move.

4.3.1 Personal Accounts

Account Type	Description
Personal Checking	Day-to-day spending and income reception; primary transaction account.
Personal Savings	Reserved capital for specific goals or emergencies.
Operations Account	Dedicated business-expense account for gig and freelance activity.
Investment Account	Long-term capital deployment in equity, funds, or portfolio assets.
Tax Reserve Account	Automated set-aside for self-employment tax obligations; quarterly payments.
Retirement Account	Long-term personal retirement savings — SEP-IRA, 401k portability, PEP.

Account Type	Description
Equity Account	Record of equity positions in external organizations or portfolio projects.
HSA Account	Health Savings Account — portable benefit, pre-tax health expense fund.
Portable Savings Account	Gig platform contribution tracking account; receives % contributions from platforms (e.g., 4% of pre-tip earnings).
Income Protection Fund	Reserves for income replacement during missed work, disability, or occupational accident.
Professional Development Account	Portable education and training fund; receives contributions from platforms or self.
Emergency Savings Account	Rapid-access emergency buffer, separate from general savings.
Crypto / Token Account	Digital asset and token holdings; on-chain transfers via crypto bridge.

4.3.2 Organizational Accounts

Account Type	Description
Operating Account	Primary revenue and expense account for the organization; multi-sig controls.
Payroll Account	Funds designated for member and worker compensation; automated distribution.
Project Account	Per-project budget pool; linked to a specific Portfolio Project.
Cooperative Treasury	Multi-sig controlled shared fund; democratic governance for disbursements.
Campaign Account	Dedicated account for an active fundraising or resource campaign.
Escrow Reserve	Collective escrow for multi-party deals; M-of-N release.
Mutual Aid Fund	Peer-support pool for member emergencies; governed by mutual aid policy.
Equity Pool	Shared equity register for member equity distribution; vesting schedules.

4.4 Wallet Types

Wallet Type	Primary Use
Personal Wallet	Day-to-day financial activity; receive income, pay expenses.
Operations Wallet	Business expense management for gig and freelance activity.
Marketplace Wallet	Buying and selling in kogi-marketplace.
Exchange Wallet	Active trading and deal execution.

Wallet Type	Primary Use
Investment Wallet	Deploying capital in equity, campaigns, and financial instruments.
Payroll Wallet	Receive compensation from contracts and gigs.
Tax Wallet	Automated tax savings and quarterly payments.
Campaign Wallet	Launch and manage fundraising campaigns.
Collective Wallet	Shared wallet for collective and cooperative member activity.
Escrow Wallet	Temporary custody of deal funds.
Crypto Wallet	Digital asset transactions; on-chain transfers.

4.5 Auto-Split Rules

A powerful Wallet feature is configurable auto-split: when income arrives in a wallet, it is automatically routed to multiple accounts according to defined percentage rules. Example: Jordan sets up 60% → Personal Checking, 20% → Operations Account, 15% → Tax Reserve, 5% → Investment Account. Every contract payment auto-applies the split atomically.

4.6 Double-Entry Ledger

The kogi-bank ledger implements double-entry bookkeeping. Every financial event creates at least two ledger entries: a debit to one account and a credit to another. The ledger is append-only and immutable — corrections are made by posting reversing entries, not by editing.

Journal Type	Entry Types
Income Journal	Client payments, marketplace proceeds, gig payouts, subscription revenue
Expense Journal	Subscriptions, tools, equipment, platform fees, professional services
Asset Journal	Equipment purchases, IP acquisitions, investment deployments
Liability Journal	Loans, deferred revenue, outstanding payables
Equity Journal	Equity issuances, cap table changes, vesting events
Escrow Journal	Escrow locks and releases from marketplace and exchange deals
Campaign Journal	Campaign contributions received and disbursed
Tax Journal	Tax reserve allocations, estimated tax payments, tax refunds
Payroll Journal	Member and worker compensation payments
Adjustment Journal	Corrections, reversals, write-offs

4.7 Escrow System

The kogi-bank Escrow System provides secure, conditional custody of funds for Marketplace and Exchange deals. Every escrow is a first-class financial entity with its own account, ledger trail, governance policies, and milestone schedule.

Escrow Lifecycle Stage	Ledger Effect
Deal Accepted	No ledger entry yet; escrow record created.
Funded	Debit: Buyer Checking → Credit: Escrow Reserve Account.
Milestone Approved	Debit: Escrow Reserve → Credit: Seller Payroll / Checking.
Dispute Raised	No movement; funds frozen; Dispute entity created.
Auto-Release	Fires N days after delivery if buyer is silent; Debit: Escrow → Credit: Seller.
Full Release	All milestones approved; Escrow account balance → 0.
Refunded	Deal cancelled; Debit: Escrow Reserve → Credit: Buyer Checking.

4.8 Campaigns & Group Economics

Campaign Type	Description
Fundraise	Donation-based funding; non-equity, non-repayable; lump sum on goal met.
Equity Crowdfunding	Fractional ownership in exchange for capital; Reg CF / Reg A+ compliance; SAFE notes supported.
Revenue Share Pool	Backers share in future revenue from a Portfolio Component.
Group Buy	Collective purchase at negotiated bulk rate; shared ownership.
Community Bond	Debt issuance from a cooperative or collective; fixed-rate repayment.
Mutual Aid Round	Community pool for member emergency support; peer reciprocity.
Pre-sale	Early commitment to purchase before launch; escrow held until delivery.
Resource Campaign	Collective gathering of non-monetary resources: compute, licenses, materials.

4.9 Portable Benefits

kogi-bank manages portable benefits for independent workers — financial benefits that travel with the worker regardless of which clients or projects they work on. Every benefit account is a Portfolio Item in the Portfolio System.

Benefit Category	Benefit Types
Health & Wellness	Health insurance, dental, vision, Health Savings Accounts (HSA)
Retirement & Savings	SEP-IRA, 401k portability, Pooled Employer Plans (PEP), emergency savings accounts

Benefit Category	Benefit Types
Paid Time Off & Income Security	Paid sick days, vacation time, income replacement for missed work
Insurance & Protection	Occupational accident insurance, disability, workers' compensation
Professional Development	Portable education accounts, training fund, certification credits
Platform Contributions	Gig platform % contributions (e.g. 4% of pre-tip earnings) to portable savings accounts; managed by benefit providers

5. Office System (kogi-office)

5.1 Overview

kogi-office is the portfolio item execution and management system. It provides the strategic management, decision management, task management, project management, and program management systems that independent workers use to execute their work. The Office integrates booking, CRM, contracts, logistics, and communication into a unified professional operations layer.

5.2 Work Management System (WMS)

The Work Management System provides planning, execution, and coordination for all work items. It uses the Work Breakdown Structure model to ensure work is traceable from strategic vision to tactical execution.

WMS Module	Primary Features
Work Dashboard	Active items summary, priority queue, velocity tracker, blocker alerts, upcoming milestones, team capacity, OKR progress, quick actions.
Backlogs	Program, project, sprint, team, and triage backlogs; MoSCoW and WSJF prioritization; dependency mapping; capacity planning.
Boards	Kanban, Agile, Resource, Priority, Program, and Custom boards; WIP limits; swimlane configurations; board policies.
Timelines	Schedule, Gantt, Calendar, Roadmap, Program Increment, Sprint, and Quarterly views; timebox model.
Work Analytics	Velocity forecasting, Monte Carlo simulations, cycle time, throughput, OKR progress, team performance dashboards.
Resource Management	Budget envelopes, resource allocation, delegation, capacity planning, TODO management.
Work CMS	Files, documents, contracts, agreements, SOPs, policies, procedures, frameworks, models — all versioned Portfolio Artifacts.

5.3 Booking & Scheduling

Feature	Description
Centralized Calendars	Unified calendar aggregating bookings, events, and milestones across office and community.
Multi-Worker Dashboards	Overview of all member schedules, availability, and booking conflicts.
Conflict Detection	Automated scheduling conflict detection; Oba AI alerts before confirmation.

Feature	Description
Event Booking & Reservations	Full lifecycle: inquiry → quote → confirm → deposit → delivery → close.
Resource Booking	Book equipment, studios, and venues alongside worker time.
Bookable Profiles	Workers and organizations can set profiles as bookable; client-facing booking widgets.

5.4 CRM & Lead Management

Feature	Description
Lead Capture Forms	Embeddable forms on profile and portfolio pages; leads flow into CRM pipeline.
Client Database	Contact info, booking history, spend, communication log, tags.
Pipeline Management	Inquiry → Qualified → Quoted → Negotiating → Booked → Retained.
Automated Follow-ups	Triggered sequences on stage changes, quote expiry, post-event completion.
Lead Scoring	CRMEngine scores probability-to-book; Oba recommends outreach timing.

5.5 Contracts & Invoicing

Feature	Description
Automated Contracts	Template contracts auto-populated from booking details; custom clauses supported.
E-Signatures	Integrated e-signature workflow via ProviderSystem (DocuSign, HelloSign).
Payment Reminders	Configurable deposit and balance due reminders via kogi-bank invoice system.
Recurring Billing	Subscription and retainer billing with automated recurring invoice generation.

5.6 Logistics & Communication

Feature	Description
Tour / Project Itinerary	Multi-stop itinerary builder with timeline view; links to bookings and assignments.
Resource Allocation	Assign equipment, vehicles, and crew to dates and times; conflict detection.

Feature	Description
Logistics Tracking	Equipment rentals, transportation, crew schedules — live status tracking.
In-App Communication	Dedicated logistics room per booking; broadcast to all crew; real-time updates.
Finance & Reporting	Real-time expense tracking vs. budget; P&L per booking; tax report generation.

6. Marketplace & Exchange

6.1 Overview

The Kogi Marketplace and Exchange constitute the economic layer of the platform — the transactional, liquidity, and matching infrastructure that connects independent workers, collectives, cooperatives, and organizations for the purpose of trading labor, skills, assets, resources, financial instruments, and portfolio components.

Design Principle: The Marketplace is where you offer what you have. The Exchange is where you trade what it's worth. Both are powered by the same Portfolio abstraction — every listing, bid, offer, and deal is ultimately a Portfolio Item with economic intent.

6.2 Marketplace Categories

Category	Description
Work	Labor, skills, and services offered for hire. Gigs, contracts, jobs, tasks, consulting engagements. Linked to Resource (Human) Portfolio Items.
Assets & Artifacts	Purchasable or licensable portfolio outputs. Playbooks, templates, guides, tools, designs, software. Linked to Asset and Artifact Portfolio Items.
Resources	Allocatable resources offered to others. Compute, licenses, equipment, capital. Linked to Resource Portfolio Items.
Campaigns	Fundraising, crowdfunding, and group economics. Equity rounds, resource pools, community funds. Linked to Program and Project Portfolio Items.

6.3 Exchange Markets

Market	What Is Traded
Labor Market (KLM)	Skills, gigs, contracts, work capacity, tasks — matching workers to opportunities via Deferred Acceptance algorithm.
Portfolio Commodities Market (KPCM)	Portfolio Items: assets, artifacts, projects, programs — listed, offered, dealt, and swapped.
Financial Instruments Market (KFIM)	Equity, revenue share notes, convertible notes, funds, tokens — order book, auctions, rounds.
Resource Market (KRM)	Compute, licenses, physical assets, capital, IP rights — spot trades, futures, leases.
Grants Marketplace	Government, foundation, platform, and community grants; discovery, application, and fund disbursement.
Microfinancing Market	Microloan origination, crowdlending, community lending pools, credit scoring via RiskEngine.

6.4 Transaction Types

Transaction Entity	Description
Bid	A buyer's competitive price offer; remains open until matched, expired, or cancelled.
Offer / Ask	A seller's price or terms publication; posted-price or declining-price mechanisms.
Deal	A negotiated transaction in progress; multi-party; stage-gated; ZOPA-engine assisted.
Request for Proposal (RFP)	Buyer posts requirements; multiple sellers submit competing proposals.
Proposal	A structured pitch or response to a request; draft → submitted → decided.
Contract	Legally binding engagement agreement with milestone payment schedule and escrow.
Gig	Single fixed deliverable; fast-track matching → accept → escrow funded in one step.
Task	Sub-unit of a Contract or Gig; allocated within an existing engagement scope.
Swap	Direct peer-to-peer exchange of portfolio items, resources, or labor — no currency required.
Retainer	Ongoing access to worker time; monthly recurring invoice and auto-payment.

6.5 Deal Room

The Deal Room is the negotiation and coordination space that every Marketplace and Exchange transaction moves through. It is equipped with:

- Conversation thread — full message history between buyer and seller
- Proposal panel — active Proposal with version history and side-by-side comparison
- Document vault — attached files, contracts, DD reports, portfolio previews
- Milestone tracker — visual milestone board with status and escrow lock amounts
- Oba AI panel — drafts terms, compares proposals, flags risk patterns, computes ZOPA range
- Escrow status widget — live view of escrow balance and release conditions from kogi-bank
- Activity log — immutable deal event history

6.6 Due Diligence System

For high-value deals in the Financial Instruments Market, kogi-exchange provides structured Due Diligence giving buyers access to governed, auditable portfolio intelligence before committing:

DD Component	Description
Portfolio Health Report	kogi-engine PortfolioHealthScore: overall score, signal breakdown, risk flags.
Financial Summary	Revenue, expenses, budget utilization, cash flow signals from kogi-bank.
Audit Trail	Full immutable EventLog from the Portfolio System for the subject component.
Risk Assessment	kogi-engine RiskEngine: risk score, risk optimization plan, red flags.
Asset Valuation	AI-computed AssetValue model with comparable transactions.
Social Proof	Community engagement, follower and subscriber count, review summary, spread analytics.

7. Spaces & Community System

7.1 Overview

The Kogi Community, Spaces and Social System is the connective social tissue of the platform. It connects independent workers to one another through shared spaces, purposeful rooms, direct and group messaging, broadcast channels, event management, and a rich notifications infrastructure — all built natively on top of the Portfolio System.

Design Philosophy: Every message, notification, room, and event is anchored to a PortfolioComponent. Social interactions generate signals that flow to kogi-engine and feed the recommendation, risk, and analytics engines.

7.2 Space Types

Space Type	Description
Community	Open or invite-only community around a topic, interest, or discipline.
Team	Collaborative working group — squads, tribes, guilds, chapters.
Organization	Formal legal or operational entity (LLC, Corp, Trust, Fund).
Collective	Informal resource-sharing group; skill pool.
Cooperative	Member-owned economic cooperative with shared profits and voting rights.
DAO / Autonomous Org	Decentralized autonomous org with on-chain or platform-native governance.
Federation	Network of linked Spaces, Collectives, or Organizations.
Project Space	Space tied to a specific portfolio project or program.
Private Studio	Personal creative or research environment.

7.3 Room Types

Room Type	Linked Entity
Direct (DM)	User ↔ User; unicast private messaging with end-to-end encryption option.
Group	Multi-user private channel; multicast up to 250 members.
Broadcast	One-to-many; only admins publish; subscribers read-only.
Space Room	Channel inside a Space: general, topic, project, announcements.
Project Room	Collaborative room tied to a specific portfolio project.
Deal Room	Negotiation space for marketplace deals and exchange proposals.

Room Type	Linked Entity
Event Room	Temporary room for an event or meeting; auto-archivable.
Governance Room	DAO and org voting and proposal discussion room.
AI Room	Chat interface with the Oba AI assistant.

7.4 Messaging Delivery Model

Delivery Mode	Description
Unicast (1:1)	One sender → one recipient. Private, end-to-end directed. DM Rooms.
Multicast (1:N)	One sender → defined set of N recipients. Group and Project Rooms.
Broadcast (1:ALL)	One authorized sender → all channel subscribers. Subscribers read-only. Announcement channels, alert channels.

7.5 Events Management

Event Type	Key Features
Meeting	Scheduled video/audio call; Zoom/Google Meet integration; recording, notes room.
Workshop	Interactive skill-building; materials library, exercise tracking, attendance certificate.
AMA (Ask Me Anything)	Open Q&A; question queue, upvote, moderation, recording.
Demo Day	Portfolio showcase; presenter slots, viewer Q&A, voting, portfolio links.
Governance Vote	Formal voting for an organization; linked to Proposal, quorum tracking, result execution.
Launch	Product/project launch event; portfolio item link, media kit, announcement broadcast.
Mastermind	Peer-to-peer knowledge sharing; round-robin structure, topic queue.
Conference	Multi-session gathering; multiple sessions, breakout rooms, speaker roles, recording.

7.6 Notifications & Alerts

Delivery Channel	Priority Routing
In-app (WebSocket)	All priorities; instant delivery when user is online.

Delivery Channel	Priority Routing
Notification Feed (KFED)	Normal and Low priority; persistent paginated inbox.
Mobile Push (FCM/APNs)	High and Critical priority; requires device token registration.
Email (SMTP)	Critical immediate; Normal as daily digest.
SMS (Twilio)	Critical only; opt-in; governance deadlines.
Third-party bridge	Slack, Discord, WhatsApp; per user preference via kogi-providers.

7.7 Organization Governance

Governance Component	Description
Proposal	Motion to take action; title, body, linked action, voting window, required quorum.
Vote	Member vote: approve, reject, or abstain; weighted by stake or 1-member-1-vote.
Quorum	Minimum participation threshold; configurable per organization type.
Resolution	Outcome of a passed proposal; may trigger automated actions via Oba agent.
Treasury Proposal	Financial action requiring approval: budget allocation, investment, distribution.
Membership Proposal	Add, remove, or change role of a member.
Amendment	Proposal to modify org bylaws, policies, or governance rules.

8. Hub — Multi-Portfolio Governance

8.1 Overview

The Hub is the multi-portfolio governance management system — the layer of the platform that enables work portfolios to be shared, collaborated on, governed, and managed across organizational boundaries. It supports decentralized organization, voting, rights management, and federated coordination.

8.2 Shared Portfolios

A Shared Portfolio is a Portfolio whose ComponentMetadata.owners[] field contains two or more distinct user or organization entity IDs. Shared portfolios support all existing lifecycle, governance, CRDT, and analytics operations, with extensions for multi-party coordination.

Shared Portfolio Type	Governance Model
Team Portfolio	Admin-controlled; leader override capability.
Organization Portfolio	Multi-sig plus governance proposal required.
Collective Portfolio	Open contribution; light moderation.
Cooperative Portfolio	Full member vote required for all significant actions.
Federation Portfolio	Inter-org governance protocol; delegated authority.
Crowdresourced Portfolio	Open or community contributors; contribution ledger; steward review.

8.3 Contribution Attribution Model

All contributions to a shared portfolio component are tracked as ContributionRecord objects appended to the component's EventLog. Each record captures contributor ID, contribution type (Labor, Capital, Asset, Knowledge, Artifact, Code, Design, Data), contribution value, attribution weight, governance status, and optional linked kogi-bank ledger entry.

The attribution weight drives distribution calculations in the CollaborationEngine, which computes real-time attribution updates, detects merge conflicts, scores collaboration health, and generates Oba recommendations for steward review queues.

8.4 Portfolio Resource Sharing

Portfolio Resource Sharing is implemented as a ResourceShare edge in the Graph system. Resources can be shared with spaces, organizations, and individual users with configurable access levels: Read, Fork, Contribute, or Manage. Shareable resource types include: Artifact, Template, Playbook, ToolBox, Contract, Design, Dataset, Notebook, Guidebook, and PlaybookSets.

8.5 Federation

Federations are cross-organization meta-entities enabling joint governance, shared portfolio management, and inter-org resource exchange between aligned organizations. Federation members can:

- Maintain a Federation Reserve — shared capital pool funded by member contributions
- Execute inter-federate transfers — resource movement between member organizations
- Launch Federation Campaigns — cross-organizational fundraising toward a shared goal
- Access consolidated reporting — federation-level financial overview aggregating all member entities
- Use federated identity — members can act on behalf of the org in federated spaces

8.6 CrossGridLink Protocol

The CrossGridLink protocol enables consent-gated data sharing between portfolios across the KLNK (Kogi Link Network). The six-phase lifecycle ensures full data sovereignty:

Phase	Description
1. Request	Source Grid requests link to Target Grid; specifies edge type and desired mirrored attributes.
2. Configure	Both parties configure consent terms; worker specifies which attributes to mirror and whether write-back is permitted.
3. Provision	ShadowCell created in host Grid; initial snapshot sync of mirrored attribute values.
4. Sync	Ongoing real-time or batch synchronization of mirrored attribute changes.
5. Writeback	Optional: Grid A can write to Grid B on consented write-back attributes.
6. Lifecycle	Consent can be revoked at any time; revocation immediately halts shadow sync; cannot be silently re-accepted.

9. Agent & Automation System

9.1 Overview

The Agent module provides portfolio automations through the Oba AI assistant and a multi-agent orchestration system. Users can connect AI agents, configure work automations, and build intelligent pipelines that execute across their portfolio.

9.2 Oba AI Assistant

Oba is the primary AI assistant for the Kogi platform, running on the Sambara agent system. Oba operates in multiple modes to provide contextual assistance across the entire platform:

Oba Mode	Description
Reactive	Responds to direct user queries in natural language; NL-to-HyperQL translation.
Proactive	Monitors portfolio signals and surfaces recommendations, alerts, and briefings.
ColumnCompletion	Auto-completes spreadsheet-style fields based on context and patterns.
SmartFilter	Translates natural language filter requests into HyperQL predicates.
FormulaSuggestion	Recommends Tier-1 formula attributes based on user's analytical intent.
BatchAction	Prepares batch operations (payroll, escrow releases, bulk status updates) for one-click approval.
HealthBriefing	Daily morning briefing: portfolio health, financial summary, opportunities, risks, action items.
NarrativeGeneration	Auto-generates narrative summaries for portfolio items; powers grant applications and client-facing summaries.
Autonomous	Executes pre-approved rules without per-action confirmation; budget-gated automations.

9.3 Connectable AI Agents

The Agent module supports connection of multiple AI providers alongside the native Oba assistant:

- Oba (native Kogi AI)
- Claude (Anthropic)
- ChatGPT (OpenAI)
- Grok (xAI)
- DeepSeek
- Gemini (Google)
- Mistral
- Perplexity

- Llama (Meta)
- Qwen (Alibaba)
- Kimi.ai
- OpenRouter
- Custom / Self-hosted

9.4 Automation Architecture

Automation Level	Description
Orchestration	Multiple workflows coordinated into automated work pipelines; cross-module execution.
Workflows	Named sequences of actions triggered by events, schedules, or conditions.
Tasks	Individual automation steps within a workflow; typed actions on portfolio items.
Processes	Repeatable process templates instantiated from playbooks.
Work Pipelines	End-to-end automated pipelines from input signal to output action.

9.5 kogi-engine Sub-Engines

The intelligence layer powering all platform modules consists of specialized engines coordinated by kogi-engine (Scala 3):

Engine	Primary Responsibility
AnalyticsEngine	Computes engagement, performance, usage analytics, and all quantitative signals.
TelemetryEngine	Real-time event streaming; component flow statistics; PlatformDataEnvelope processing.
RecommendationEngine	Collaborative and content-based filtering; personalized discovery; cold-start handling.
PersonalizationEngine	Persona lifecycle management; A/B experiments; UCB1 multi-armed bandit; content delivery config.
MatchEngine	Multi-dimensional worker-to-opportunity, investor-to-campaign, resource-to-project matching.
GraphEngine	Dependency graph, critical path analysis, betweenness centrality, community detection.
RiskEngine	PortfolioHealthScore; RiskOptimizationPlan; anomaly detection; counterparty risk scoring.
OptimizationEngine	Resource allocation optimization; capacity planning; bottleneck detection; flow efficiency.
SearchEngine	Full-text and semantic search across all component types; personalized re-ranking.

Engine	Primary Responsibility
AllocationEngine	Deferred Acceptance algorithm for labor markets; capital allocation; attention markets.
IncentiveEngine	Reputation scoring; KogiPoints accounting; streak tracking; Shapley value distribution.
CollaborationEngine	Shared portfolio coordination; contribution attribution; conflict resolution; health scoring.
BenefitsEngine	Benefits eligibility scoring; coverage optimization; gap analysis; contribution forecasting.
GrantEngine	Grant matching; eligibility scoring; impact tracking; community grant pool management.
CRMEngine	Lead scoring; pipeline management; follow-up automation; retention analytics.
BookingEngine	Scheduling optimization; conflict detection; resource allocation; utilization analytics.
LogisticsEngine	Itinerary planning; resource routing; crew scheduling; expense tracking.
DataStreamingEngine	Real-time data export; BI connector feeds; event bus for third-party integrations.

10. Mechanism Design & Game System (kogi-game)

10.1 Overview

kogi-game is the mechanism design and game engine of the Kogi Platform — the mathematical and computational substrate that governs how resources, capital, labor, portfolio components, and financial instruments are discovered, matched, priced, allocated, exchanged, and incentivized.

Design Principle: Mechanism design is the engineering of economic games. kogi-game designs rules that align individual incentives with platform-wide efficiency for every interaction — from a single gig bid to a multi-million dollar equity round.

10.2 Game Types

Game Type	Mechanism
Assignment Game	Worker-to-gig matching via Deferred Acceptance (Gale-Shapley) with multi-dimensional scoring.
Double Auction	Exchange market: Continuous Double Auction with price-time priority order book.
Vickrey Auction (VCG)	Single-item high-value sales: second-price sealed-bid; incentive-compatible.
Reverse Auction / RFP	Score-based sealed-bid reverse auction with price+quality scoring function.
Negotiated Exchange	Alternating-offer Rubinstein bargaining with AI-assisted ZOPA computation.
Cooperative Game	Shapley Value computation for fair contribution-based distribution in collectives.
All-or-Nothing Campaign	Assurance contract / threshold public goods game; auto-refund on failure.
Reputation Game	Repeated game with Folk Theorem equilibrium; credible threat of reputation penalties.

10.3 Reputation System

Every user and organizational entity has a multi-dimensional ReputationScore computed as:
 $R(u) = 0.30 \times \text{Delivery Reliability} + 0.25 \times \text{Work Quality} + 0.20 \times \text{Communication} + 0.15 \times \text{Community Contribution} + 0.10 \times \text{Financial Integrity}$

Tier	Score Range	Benefits	Access
Elite	90–100	Featured placement × 1.5; fees –20%; early	All marketplace, exchange, and campaign types

Tier	Score Range	Benefits	Access
		access to beta; priority dispute resolution	
Trusted	75–89	Featured placement × 1.2; fees –10%; verified badge on all listings	All marketplace and exchange types
Established	55–74	Standard placement and fees; verified identity badge	All standard marketplace and exchange types
Developing	35–54	Standard placement and fees; no promoted listing access	Standard gigs, proposals, and campaigns only
New	0–34	Cold-start tier; guided onboarding; limited deal sizes; escrow mandatory	Simple gigs and low-value transactions only

10.4 KogiPoints Incentive System

KogiPoints (KP) are the platform's internal incentive currency — non-monetary credits earned through value-creating behaviors and redeemable against platform fees, promoted listing boosts, and premium features. KP is non-transferable between users and may not function as currency.

Earning Event	KP Awarded
Complete a gig / contract (seller)	50–500 KP (proportional to contract value)
Receive 5-star rating	25 KP
Publish a portfolio component to marketplace	75 KP
Successful referral (new user completes first deal)	250 KP
Complete onboarding profile (skills + portfolio + availability)	100 KP one-time
Participate in cooperative governance vote	10 KP
Mentor a Newcomer (guided interaction)	50 KP

11. Resource Management System (KOGI-RMS)

11.1 Overview

KOGI-RMS is the unified layer for discovering, organizing, allocating, governing, and tracking every resource on the Kogi platform. A resource is any entity — structured or unstructured, human or non-human, tangible or intangible — that has value, can be acted upon, and can be associated with a Portfolio component.

11.2 Resource Type Taxonomy

Domain	Resource Categories
Portfolio & Work	Portfolios, programs, projects, assets, artifacts, binders, journals, books, dossiers, folders, documents, files, directories, registries, archives
Planning & Time	Timelines, schedules, roadmaps, calendars, Gantt charts, timeboxes, milestones, roadblocks
Work Execution	Boards, epics, stories, features, tasks, work packages, WBS items, initiatives, strategies, themes, goals, OKRs
Commerce & Contracts	Gigs, bookings, consultations, jobs, contracts, offers, deals, requests, proposals, bids, investments, orders, invoices
Capital & Physical	Capital, financial assets, labor, land, estates, real estate, equipment, inventory, IP, data, credits
Financial Instruments	Equity, liquidity, debt, taxes, HSA, retirement, income protection, portable savings, revenue share
Contributions	Labor, skills, financial, support, advertising, marketing, promotions, endorsements, donations, investments
Users & Roles	Members, contributors, donors, investors, subscribers, followers, watchers, owners, editors, stewards

11.3 Worker Types

Worker Type	Platform Capabilities
Contractor	Gig listings, project contracts, invoicing, bookings, multi-client CRM. 1099/self-employment tax; HSA eligible.
Consultant	Consultation packages, retainer billing, proposals, thought leadership. Full portable benefits suite.
Gig Worker	Gig claims, rapid booking, task queues, platform contribution tracking. Portable savings account.
Freelancer	Service listings, project portfolios, client management, milestone billing.

Worker Type	Platform Capabilities
Entrepreneur	Company portfolios, equity crowdfunding, team management, cap table. Full equity and tax toolset.
Micropreneur	Solo product listings, subscriptions, microsite, community monetization. Simplified tax tracking.
Coach	Consultation booking, program creation, community spaces, certification. CRM + recurring billing.
Employee	Organization portfolio access, team boards, payroll via kogi-bank.
Officer	Org governance, signatory authority, multi-sig approvals, cap table management.

12. Technical Architecture

12.1 Service Architecture

Service	Technology Stack
Portfolio System (core)	Rust — in-memory domain engine; PortfolioSystem struct; CRDT log; EventLog; federation
kogi-bank	Go (REST/gRPC API) + Rust (escrow state machine); PostgreSQL (SERIALIZABLE for ledger); Kafka
kogi-marketplace / kogi-exchange	Go (REST/GraphQL); Rust (core modules); PostgreSQL + Redis + Kafka
kogi-community / kogi-spaces	Go (10 services: community, room, message, notification, event, feed, social graph, post, match, analytics)
kogi-office / kogi-wms	Go (API + WBS engine); Scala 3 (planning analytics via kogi-engine)
kogi-engine (AI intelligence)	Scala 3 / JVM (all sub-engines); gRPC server port 9100; Kafka consumer; Akka Streams
kogi-game	Go (primary service) + Rust (core mechanism engine via FFI); Scala 3 sub-engines embedded in kogi-engine
Hypergrid substrate	Rust (cell store, CRDT merge engine, federation manager); PostgreSQL + Redis Cluster + Kafka + ClickHouse + Meilisearch

12.2 Data Storage

Store	Usage
PostgreSQL	Primary persistence: all domain entities, ledger (SERIALIZABLE), CRDT log, EventLog, federation peers
Redis Cluster	Hot-path cache: cell cache (TTL 30s), order book state, active escrow, session deal context, presence
Apache Kafka	Event streaming: all domain events, AI feature streaming, federation delta sync, analytics pipeline
ClickHouse	Analytics: DimFold queries > 500K rows; time-series aggregations; N=3/4 dimensional analysis
Meilisearch / Typesense	Full-text search: listing search, portfolio component search, user discovery
S3 / MinIO	Object storage: snapshots, archives, media files, event recordings, EventLog S3 archive
SQLite (kogi-local)	Local/desktop client cache: offline message queue, local feed snapshot, notification inbox cache

12.3 Communication Protocols

Protocol	Usage
REST / GraphQL	External API surface for all consumer clients (web, mobile, third-party integrations)
gRPC (internal)	Service-to-service communication; kogi-engine EngineGrpcServer on port 9100
WebSocket	Real-time delivery: Deal Room, room messages, presence indicators, live escrow status
Kafka (streaming)	Event bus, AI feature streaming, federation delta sync, analytics pipeline, audit log

12.4 Client Architecture

Client	Technology
Web Client	Angular + TypeScript; WebSocket for real-time Deal Rooms and collaboration
Mobile (Android)	Kotlin; push notifications for transactions and deal events
Mobile (iOS)	Swift/iOS; push notifications via APNs
Desktop (kogi-local)	Electron or native; SQLite local cache; offline-capable
API / CLI	REST + gRPC; Bazel monorepo build system

12.5 Security & Compliance

- PCI DSS compliance — all card data handled by Stripe; kogi-bank stores no raw card numbers
- All transactions require explicit user approval — no AI-initiated fund movements without user confirmation
- Multi-factor authentication required for transactions above configurable thresholds
- Encryption at rest (AES-256) for all financial data; TLS 1.3 in transit
- Audit trail via immutable ledger + Portfolio System EventLog — every action logged with actor, timestamp, and result
- KYC / AML screening for accounts above transaction thresholds via third-party provider integration
- GDPR / CCPA right-to-be-forgotten — personal data anonymization on account deletion; ledger entries preserved for regulatory purposes
- Role-based access control — PermissionTier hierarchy governs all account, wallet, and portfolio operations
- Append-only ledger enforced at three layers: application code, PostgreSQL RLS, and periodic Merkle tree verification

13. Platform Integrations

13.1 Third-Party Payment Rails

Rail	Use Case
Stripe	Client card payments, marketplace payouts, subscription billing, identity verification
ACH (via Plaid)	Bank-to-bank transfers, payroll, external account linking
PayPal / Venmo	Small payments, contractor payouts, cross-platform transfers
Ethereum / ERC-20	Token payments, crypto payouts, DeFi activity, on-chain escrow (Phase 4)
Coinbase Commerce	Crypto invoice acceptance
SEPA	European bank transfers
Wire Transfer	Large international payments

13.2 Work & Marketplace Integrations

Category	Platforms
Freelance Markets	Upwork, Fiverr, Toptal, Freelancer — profile import, listing sync, order bridging
Portfolio / Creative	Behance, Dribbble, GitHub, GitLab — portfolio import, project showcase sync
Job Markets	LinkedIn, Indeed, AngelList — job listing sync, profile export
Collaboration	Slack, Notion, Google Workspace, Microsoft Teams — project room linking, file sync
Commerce	Gumroad, Shopify, Stripe — asset sales routing, payment bridging
Investment / Trading	Robinhood, SoFi, Interactive Brokers — portfolio linking, position sync
Equity Platforms	AngelList, Republic, Wefunder — round sync, cap table bridge
Crypto / Web3	Coinbase, Ethereum, Polygon, Solana — token bridge, wallet connect, NFT

13.3 Social Platform Integrations

Platform	Capabilities
Facebook / Instagram	Cross-post content, import followers, sync events
YouTube	Embed video, sync channel activity, cross-publish; event recording upload

Platform	Capabilities
LinkedIn	Professional profile sync, job post distribution, network import
Twitter / X	Cross-post announcements, track mentions, import audience
Slack / Discord	Community server bridging, role sync, notification relay
WhatsApp	Broadcast messages, customer communication bridge, KNTF relay
Zoom / Google Meet	Event room video calls, meeting scheduling integration
Gmail	Email-to-room bridge, DM-to-email relay, notification email delivery
Twilio	SMS notification relay, 2FA, event reminders (critical alerts)

14. Platform Economics & Fee Design

14.1 Fee Architecture

Platform fees are designed to sustain operations and curate quality while not deterring participation. Fees scale with transaction value (not fixed), are lower for repeat participants (loyalty tiers), and a portion of fee revenue is recycled into liquidity incentives.

Transaction Type	Fee Structure	Rationale	Loyalty Discount
Gig / Contract (seller)	8% → 4% at Elite tier	Covers platform services + escrow; incentivizes quality	–1% per tier (8%/7%/6%/4%)
Gig / Contract (buyer)	2% of contract value	Low buyer fee to maintain demand-side participation	–0.5% at Trusted+; waived at Elite
Marketplace listing sale	6% of sale price	Platform facilitation + discovery services	–1.5% at Elite tier
Exchange trade	0.5% of trade value (both sides)	Order book maintenance, settlement, anti-fraud	50% reduction for KP redemption
Campaign (success fee)	5% of successfully raised funds	Campaign hosting + disbursement services	–1% for cooperative entities
Campaign (equity)	3% + 0.5% carried interest	Reflects higher-value financial facilitation	Fixed (regulatory reasons)
Subscription listing	1% of recurring revenue	Ongoing platform services for subscription management	Waived at Elite tier

14.2 Fee Redistribution

Pool	Allocation
Liquidity Incentive Pool	15% of fee revenue — KP bonuses to participants in thin markets and new categories.
Newcomer Subsidy Pool	5% of fee revenue — reduced fees and matching bonuses for users completing their first 3 transactions.
Community Fund	5% of fee revenue — administered by platform governance council; grants for open community contributions.
Cooperative Bonus	5% of fee revenue — additional fee discounts for formally incorporated cooperative entities.

15. Initiatives & Special Programs

15.1 Overview

Initiatives are coordinated platform-level programs that workers and organizations can enroll in to access collective benefits, capital, and resources. Initiatives integrate across the Dashboard, Bank, Community, and Hub modules.

15.2 Initiative Catalog

Initiative	Description
Portable Benefits	Coordinate worker groups for pooled benefits purchasing power and Pooled Employer Plan (PEP) enrollment. BenefitsEngine provides eligibility scoring, coverage optimization, and gap analysis.
Grants & Microfinancing	Grant discovery, application, community lending pools, microloan origination. GrantEngine scores grant matches by eligibility, portfolio strength, and historical award data.
Group Economics	Cooperative treasury, revenue sharing, group investment pools, mutual aid funds. Revenue distribution engine computes Shapley Value-based patronage distributions.
Equity Crowdfunding	Reg CF/A+ campaigns, revenue share, SAFE notes, cap table management. CrowdfundingEngine scores campaigns, matches investors, models equity, and checks compliance.
Crowdresourcing	Community-sourced labor, capital, asset, and knowledge contributions with full attribution tracking and configurable reward rules.
Shared Portfolios	Multi-user, multi-org portfolio management with contribution attribution and governance. CollaborationEngine coordinates concurrent edits and health scoring.
Portfolio Resource Sharing	Share artifacts, templates, tools, playbooks, and data within community contexts. ResourceShare graph edges with configurable access levels.

16. Open Items & Roadmap

16.1 Near-Term Open Items

Item	Description
Persistence Layer (Portfolio System)	Portfolio System is currently in-memory. Implement pluggable PortfolioStore backend with PostgreSQL + Redis + S3 implementations.
CRDT Status Field Merge	apply_crdt_op() currently skips status mutations. Implement Lattice CRDT for lifecycle status fields before multi-node deployments.
Ledger Snapshot Optimization	Balance recomputation from full ledger history will slow for long-lived accounts. Implement periodic balance snapshots with verified snapshot + incremental replay.
Saga Compensation Protocol	Distributed transactions spanning kogi-bank, kogi-exchange, and kogi-portfolio need fully specified Saga compensation maps for escrow, deal, and campaign settlement.
KYC / AML Integration	Select and integrate a third-party KYC/AML provider (Stripe Identity, Persona, Onfido) via kogi-providers for accounts above defined thresholds.
AI Finance Agent Writeback Protocol	Define typed writeback protocol for Finance Agent to create Invoice drafts, schedule transactions, and append financial governance notes without bypassing PermissionTier checks.
Portable Benefits Provider Integration	Define ProviderSystem integration pattern for benefit account management with third-party retirement, health, and insurance providers.
SearchEngine Warm-up	Wire on_component_created / on_component_updated plugin hooks to populate Meilisearch listing index.
Combinatorial Auction Solver	Implement LP relaxation + branch-and-bound for small N; greedy approximation for large N; bid complexity limit for tractability.

16.2 Phase 4 Considerations

- On-chain escrow smart contracts — self-executing escrow for crypto/token deals without custodial counterparty risk
- Decentralized finance (DeFi) integration — yield on idle campaign and reserve accounts via DeFi protocols
- Autonomous financial agent — Oba acts on pre-approved rules: auto-pay recurring bills, auto-invest above balance threshold
- Cross-platform marketplace federation — federated listing discovery across ShangoOS platforms (Kogi, Ume, Qala)
- Real-world asset (RWA) tokenization — platform infrastructure to tokenize and trade physical assets on-chain
- Cross-platform reputation portability — export Kogi reputation signals to external platforms via verifiable credential standard
- Prediction markets — workers and investors bet on project outcomes; outcomes calibrate RiskEngine priors

- AI-autonomous market making — Oba-class agents post algorithmic bids/offers to thin markets

Appendix A: Glossary

Term	Definition
Portfolio	The root domain concept of Kogi — the complete, unified record of an independent worker's professional and economic existence. The master hyperspreadsheet IS the Portfolio.
HyperRow	A row in a Hypercube; the basic entity unit in the Hypergrid substrate.
Hypercube	An N-dimensional data store whose primary axes are entity (D_1) and property (D_2), with optional additional axes for time, jurisdiction, or segment.
CRDT	Conflict-free Replicated Data Type — distributed data structure enabling safe concurrent edits across portfolio nodes and federated deployments.
EventLog	Append-only, immutable audit history attached to every Portfolio Component recording all mutations with actor, timestamp, and delta.
VectorClock	Per-component distributed logical clock used for causal ordering of concurrent mutations across federation nodes.
KLNK	Kogi Link Network — the economic graph of interconnected worker portfolios via CrossGridLink edges.
CrossGridLink	A consent-gated graph edge connecting entities across different Portfolio Systems (Grids); enables attribute mirroring with worker data sovereignty.
ShadowCell	A mirrored attribute value in Grid A that reflects a consented attribute from Grid B, synced via the CrossGridLink protocol.
Zawadi	The basic unit of transaction on the Kogi platform; translates to/from liquidity, equity, securities, credits, debts, and digital currency.
Sundiata Coin	The principal cryptocurrency for the Kogi platform on the Sundiata distributed ledger.
Oba	The Kogi platform AI assistant; surfaces recommendations, analytics summaries, governance alerts, and optimization suggestions across all modules.
Linktree	A user's 'personal work IP address/router' — the interface for all external communications, accounts, integration toolchains, and RBAC connections.
KogiPoints (KP)	Non-monetary platform incentive credits earned through value-creating behaviors; redeemable against platform fees and listing boosts.
Portable Benefits	Worker-centered benefits that remain with the individual rather than being tied to a single employer or gig platform.
Cooperative Treasury	A multi-sig controlled shared fund for a cooperative's operations, payroll, and investment; governed by democratic member vote.

Term	Definition
Shapley Value	The unique fair allocation that assigns each cooperative member their marginal contribution averaged over all possible coalition orderings.
ZOPA	Zone of Possible Agreement — the range of prices/terms where both parties' reservation values overlap; computed by the Deal Room ZOPA Engine.
Timebox	A fixed time period bounding committed work items: sprint, PI, quarter, or custom duration.
Persona	An interest and identity classification assigned to a user that shapes recommendation, matching, UI configuration, and community cohesion.